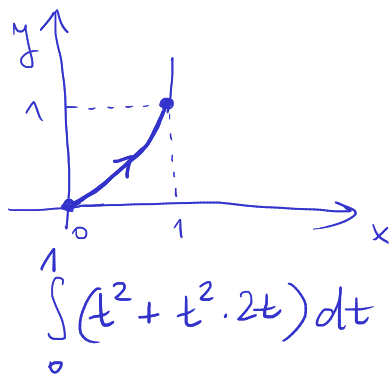


Bonus 5 - RESENU!

1)



$$x=t \Rightarrow dx=dt$$

$$y=t^2 \Rightarrow dy=2t dt \quad , t \in [0,1]$$

$$\int_0^1 (t^2 + t^2 \cdot 2t) dt = \left[\frac{t^3}{3} + 2 \cdot \frac{t^4}{4} \right]_0^1 = \frac{1}{3} + \frac{1}{2} = \underline{\underline{\frac{5}{6}}}$$

$$\begin{aligned} 2) \quad P &= 3x^2 + 2y^2 - 1 \Rightarrow \frac{\partial P}{\partial y} = 4y \\ Q &= 4xy \Rightarrow \frac{\partial Q}{\partial x} = 4y \end{aligned} \quad \Rightarrow \text{NEZAVISNI!}$$

$$\Phi = \int (3x^2 + 2y^2 - 1) dx = \underline{x^3} + \underline{2xy^2} - \underline{x} + C_1(y)$$

$$\Phi = \int 4xy dy = \underline{2xy^2} + \underline{C_2(x)}$$

$$\Rightarrow \underline{\Phi = 2xy^2 + x^3 - x}$$

$$\Phi(2,1) - \Phi(0,0) = 4 + 8 - 2 - 0 = \underline{\underline{10}}$$